

1. Question 1: Event Management System

Consider an event management system. The event is defined by

Event(eventId, title, eventType, location, eventCharge, status (Completed/Ongoing/Booked)).

The event types are IndoorEvent (GST is applicable as separate service charges) and OutdoorEvent.

(The outdoor event may be executive or non-executive, the charges are applied based on these two parameters and defined as constants, and GST is applicable as separate service charges.)

There is also an eventManager.

The OutdoorEvent is charged per hour, and IndoorEvent is charged per day.

Only booked events can be cancelled

Ongoing or completed events cannot be cancelled.

Every event must be assigned a manager, though a manager may not be assigned to any event.

- ✓ Create a few indoor and outdoor events.
- ✓ Apply runtime polymorphism using Event reference variables to compute charges.
- ✓ Print event details and list of cancelled events.
- ✓ Print total revenue generated from organizing all events.
- ✓ Identify and manage runtime errors using user-defined exception handling in Java.

2. Question 2: Loan Processing System

Consider a Loan Processing System. Each loan is defined by

Loan(loanId, applicantName, loanType, principalAmount, interestRate, durationInYears, loanStatus (Approved/Pending/Rejected)).

The loan types include HomeLoan and VehicleLoan.

For HomeLoan, an additional processing fee is charged (defined as a constant), and GST is applicable as separate service charges.

For VehicleLoan, if the applicant is a corporate customer, an additional corporate service tax is applied (defined as a constant). GST is applicable to all loan types.

- ✓ Loans can only be approved or rejected if their status is Pending.
- ✓ Rejected or already approved loans cannot be modified.
- ✓ Every loan is assigned to a LoanManager. A manager may exist without managing any loan.
- ✓ Create a few loan applications for both types of loans.

- ✓ Use runtime polymorphism via Loan reference variables to compute total repayment amount (principal + interest + service charges).
- ✓ Print loan details and statuses.
- ✓ Try to approve and reject loans as per the business rules.
- ✓ Compute and print the total revenue from all loans (interest + service charges).
- ✓ Identify and manage runtime exceptions using user-defined exception handling.

3. Question 3: Course Registration System

Consider a Course Registration System. Each course is defined by

Course(courseId, title, courseType, durationInWeeks, baseFee, status
(Available/Started/Completed)).

The course types include TechnicalCourse and SoftSkillCourse.

TechnicalCourse includes an additional lab fee (defined as a constant) and GST is applied as a separate charge.

SoftSkillCourse includes a trainer fee if the trainer is external (boolean flag), and GST is applicable as a separate service charge

Courses are charged per week, and the final charge is calculated accordingly.

- ✓ Only Available courses can be cancelled.
- ✓ Started or Completed courses cannot be cancelled.
- ✓ Every course is assigned to a Trainer. A trainer may exist without handling any course.
- ✓ Create a few technical and soft skill courses.
- ✓ Use runtime polymorphism using Course reference to calculate final charges.
- ✓ Print all course details and cancelled courses.
- ✓ Print the total revenue generated from active (non-cancelled) courses.
- ✓ Handle runtime errors (like cancelling a non-cancellable course) using user-defined exception handling in Java.